Agile Development Process and DevOps Lab

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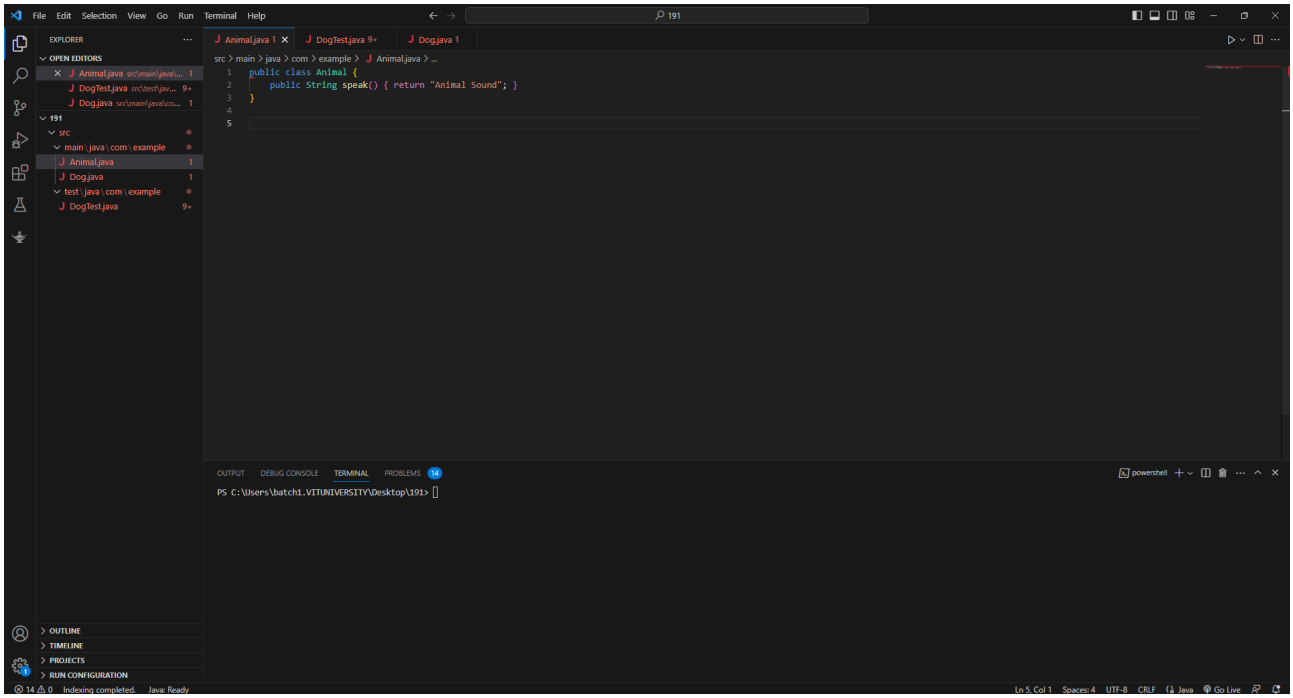
13-4-2026
VL2025260502388 - ISWE406P
Instructions:
1. COPY GITHUB link
2. Copy the Pipeline script.
3. Screenshots of the stage view and console output.
4. Complete the program as per the allotted questions.

QUESTIONS
1. For the java application automate the continuous integration process using Jenkins.
🕐 Create a Maven-based application adhering to the standard Maven directory structure.
🕐 Write the code in Visual Studio and push it to your GITHUB repository.
🕐 Modify the code and commit the updates back in GITHUB.
🕐 Using Jenkins configure the job as pipeline and create the stages for the Continuous Integration process.
🕐 Checkout GITHUB
🕐 Test using JUNIT testing.
🕐 Build using Maven
🕐 Create Docker image
🕐 Push to Docker Hub
b. To implement the inheritance concepts.



File Edit Selection View Go Run Terminal Help

EXPLORER

OPEN EDITORS

Animal.java src\main\java\... 1

DogTest.java src\test\jav... 9+

Dog.java src\main\java\co... 1

191

src

main \ java \ com \ example

Animal.java 1

Dog.java 1

test \ java \ com \ example

DogTest.java 9+

Animal.java 1 | DogTest.java 9+ | Dog.java 1

src > test > java > com > example > DogTest.java > ...

```java
import static org.junit.jupiter.api.Assertions.assertEquals;
import org.junit.jupiter.api.Test;
public class DogTest {
    @Test
    public void testSpeak() {
        Dog dog = new Dog();
        assertEquals("Woof", dog.speak());
    }
}
```

OUTPUT DEBUG CONSOLE TERMINAL PROBLEMS 14

powershell

```
PS C:\Users\batch1.VITUNIVERSITY\Desktop\191>
```

OUTLINE

TIMELINE

PROJECTS

RUN CONFIGURATION

14 0 Indexing completed. Java: Ready

Ln 10, Col 1 Spaces: 4 UTF-8 CRLF Java Go Live

File Edit Selection View Go Run Terminal Help

EXPLORER

OPEN EDITORS

Animal.java src\main\java\... 1

DogTest.java src\test\jav... 9+

Dog.java src\main\java\co... 1

191

src

main \ java \ com \ example

Animal.java 1

Dog.java 1

test \ java \ com \ example

DogTest.java 9+

Animal.java 1 | DogTest.java 9+ | Dog.java 1

src > main > java > com > example > Dog.java > Language Support for Java(TM) by Red Hat > Dog

```java
public class Dog extends Animal {
    @Override
    public String speak() { return "Woof"; }
}
```

OUTPUT DEBUG CONSOLE TERMINAL PROBLEMS 14

powershell

```
PS C:\Users\batch1.VITUNIVERSITY\Desktop\191>
```

OUTLINE

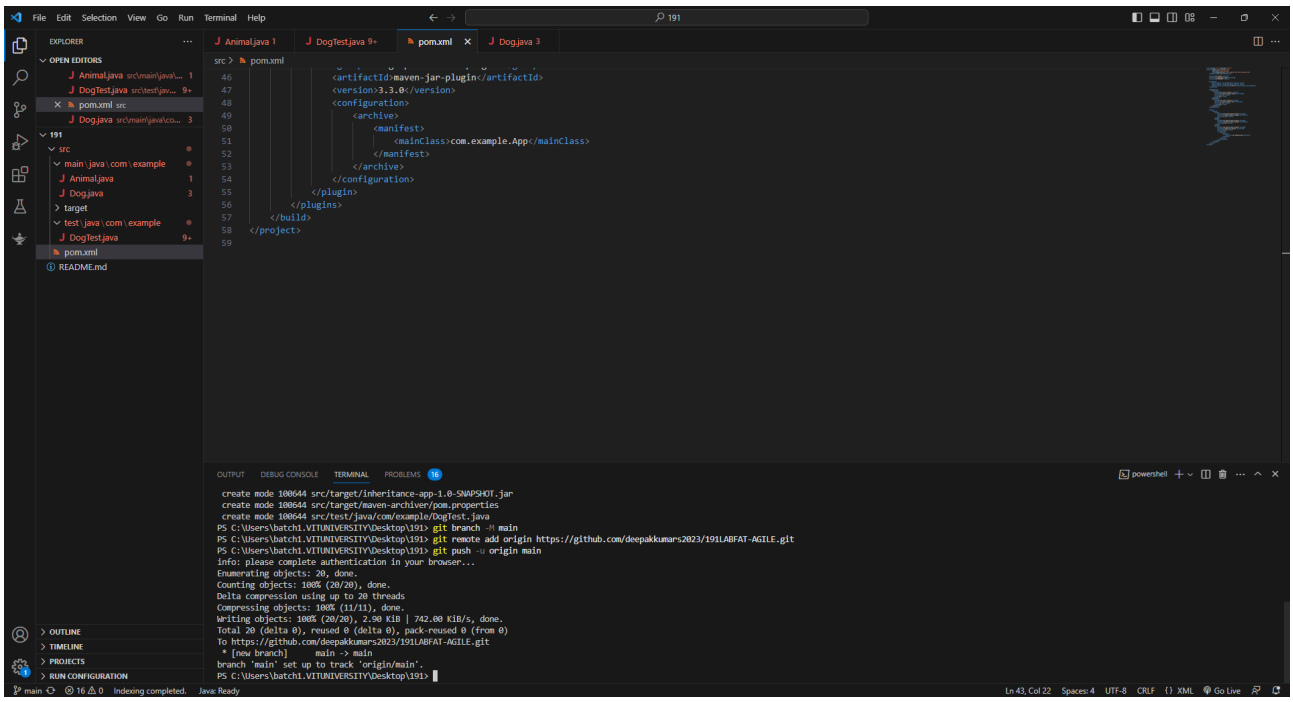
TIMELINE

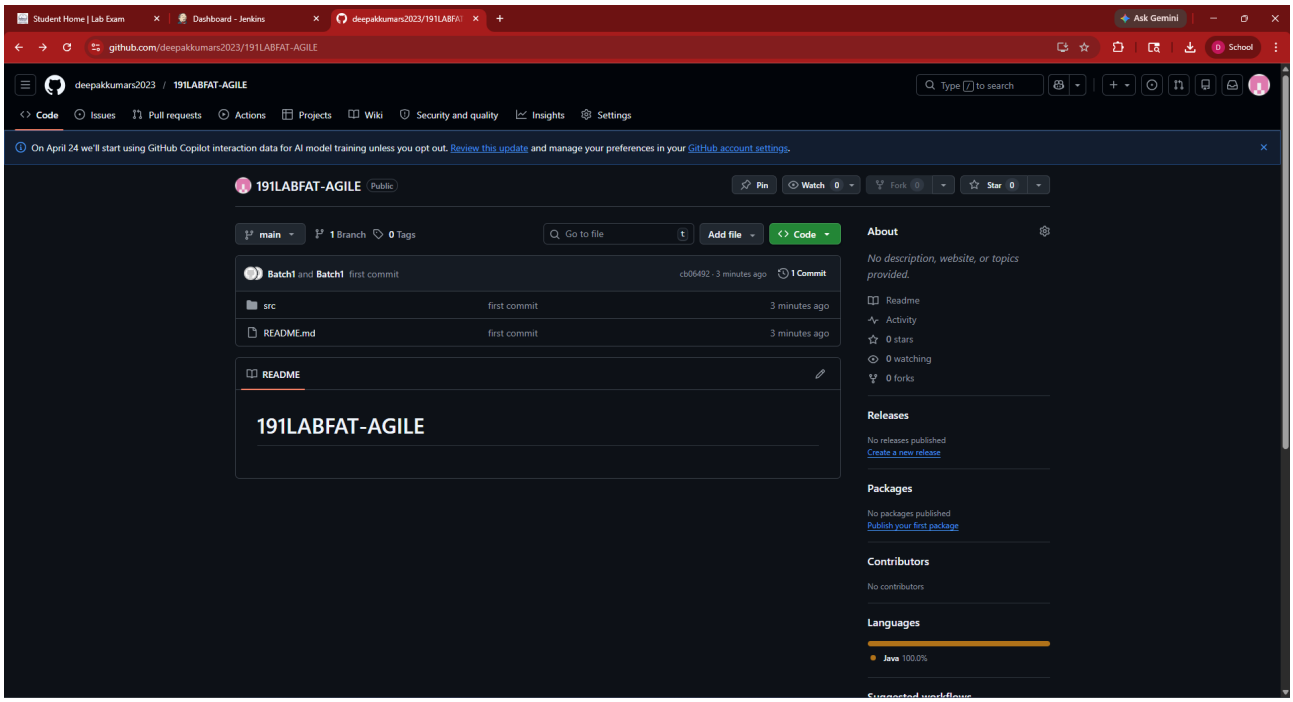
PROJECTS

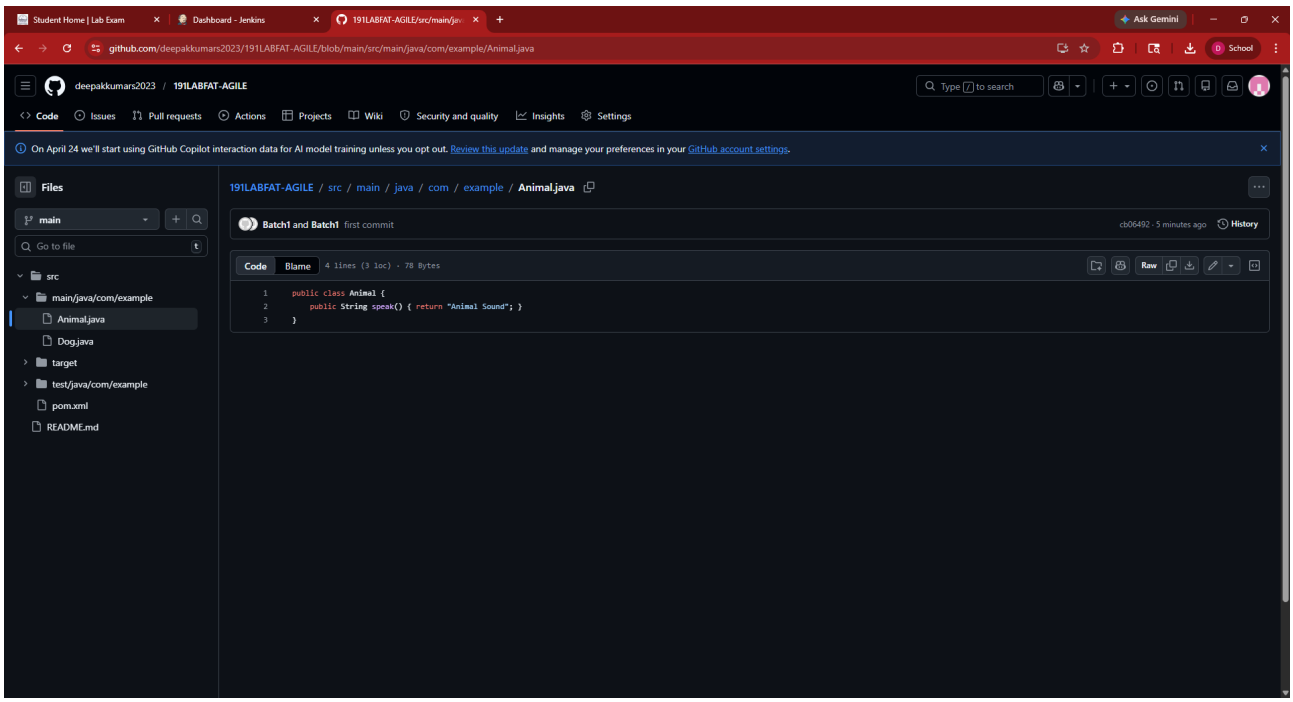
RUN CONFIGURATION

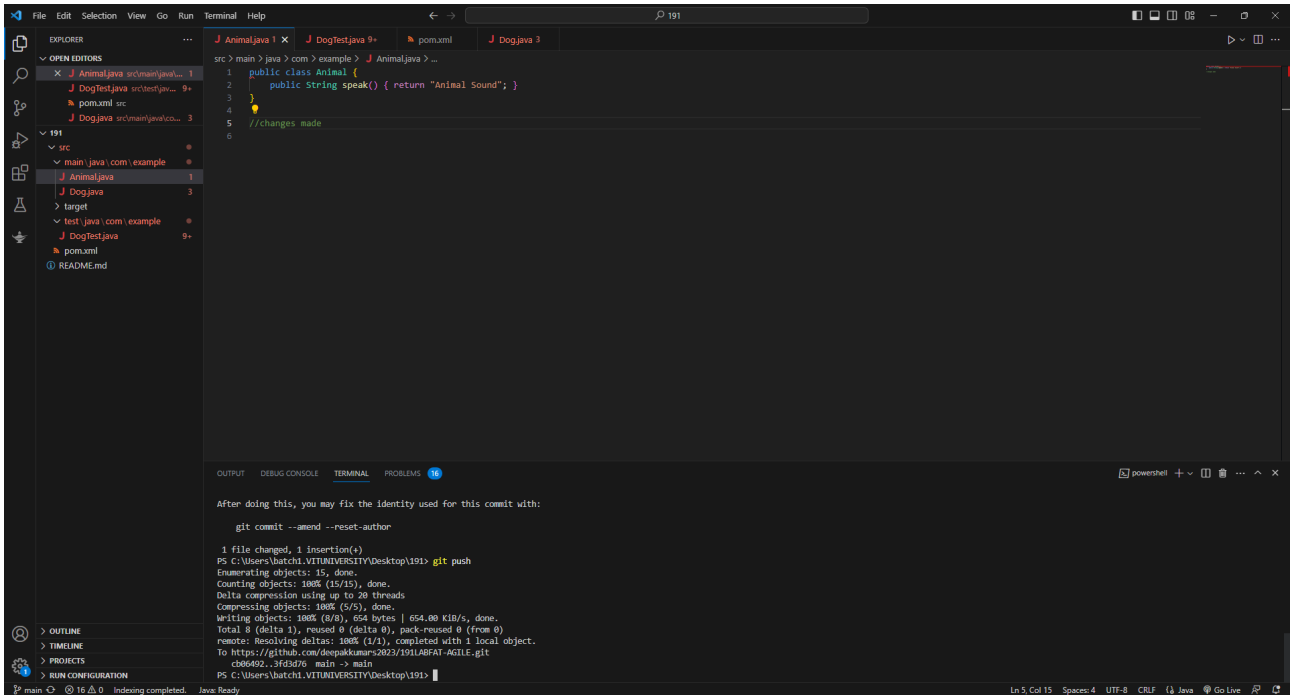
14 0 Indexing completed. Java: Ready

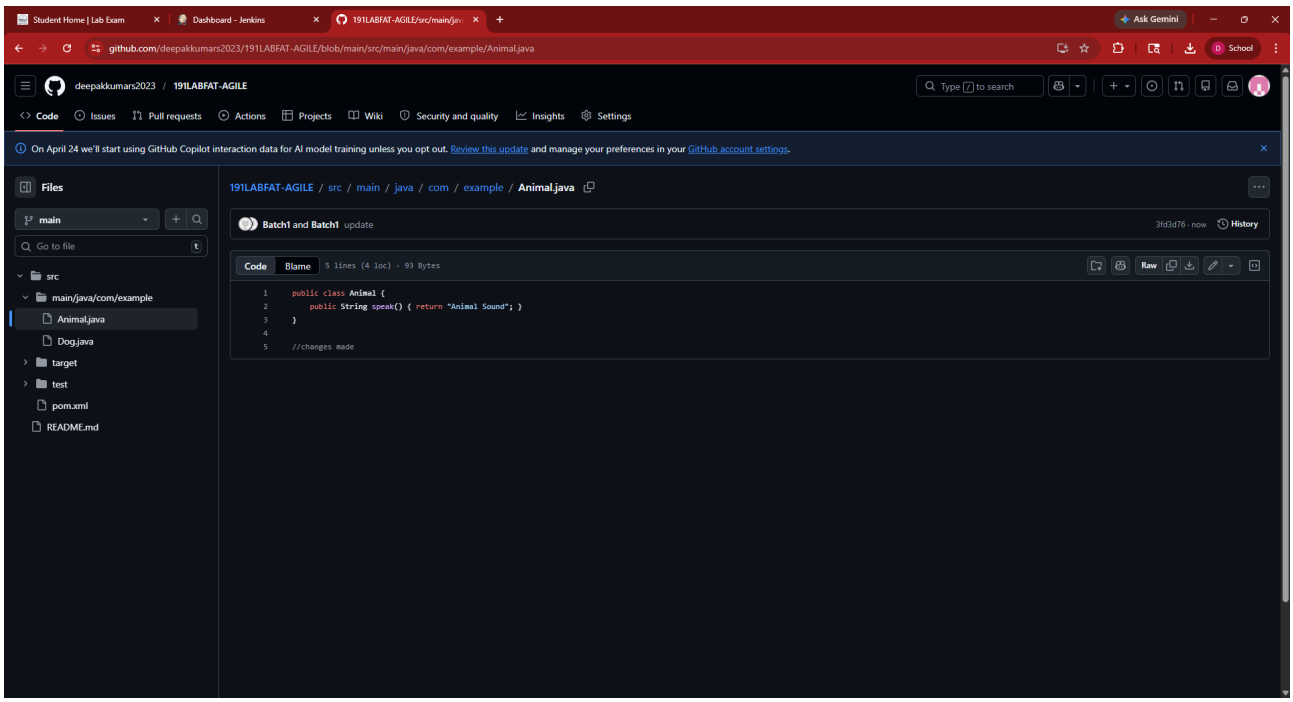
Ln 1, Col 35 Spaces: 4 UTF-8 CRLF Java Go Live

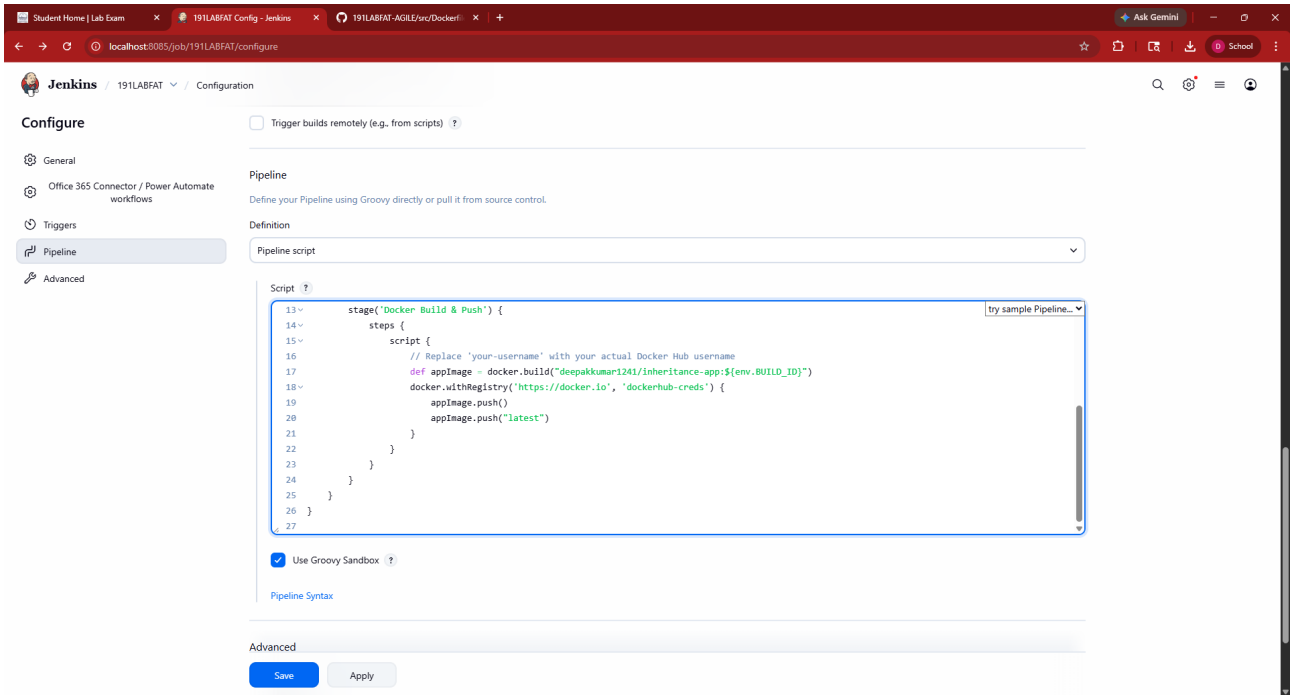

File Edit Selection View Go Run Terminal Help
191
EXPLORER
OPEN EDITORS
Animal.java 1
DogTest.java 9+
pom.xml
Dog.java 3
src > pom.xml
<artifactId>maven-jar-plugin</artifactId>
<version>3.3.0</version>
<configuration>
<archive>
<manifest>
<mainClass>com.example.App</mainClass>
</manifest>
</archive>
</configuration>
</plugin>
</plugins>
</build>
</project>
OUTPUT DEBUG CONSOLE TERMINAL PROBLEMS 16
powershell
create mode 100644 src/target/inheritance-app-1.0-SNAPSHOT.jar
create mode 100644 src/target/maven-archiver/pom.properties
create mode 100644 src/test/java/com/example/DogTest.java
PS C:\Users\batch1.VITUNIVERSITY\Desktop\191> git branch -M main
PS C:\Users\batch1.VITUNIVERSITY\Desktop\191> git remote add origin https://github.com/deepakkumars2023/191LABFAT-AGILE.git
PS C:\Users\batch1.VITUNIVERSITY\Desktop\191> git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 20, done.
Counting objects: 100% (20/20), done.
Delta compression using up to 20 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (20/20), 2.90 KiB | 742.00 KiB/s, done.
Total 20 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/deepakkumars2023/191LABFAT-AGILE.git
* [new branch] main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\batch1.VITUNIVERSITY\Desktop\191>
OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION
main
Indexing completed.
Java: Ready
Ln 43, Col 22 Spaces: 4 UTF-8 CRLF XML Go Live

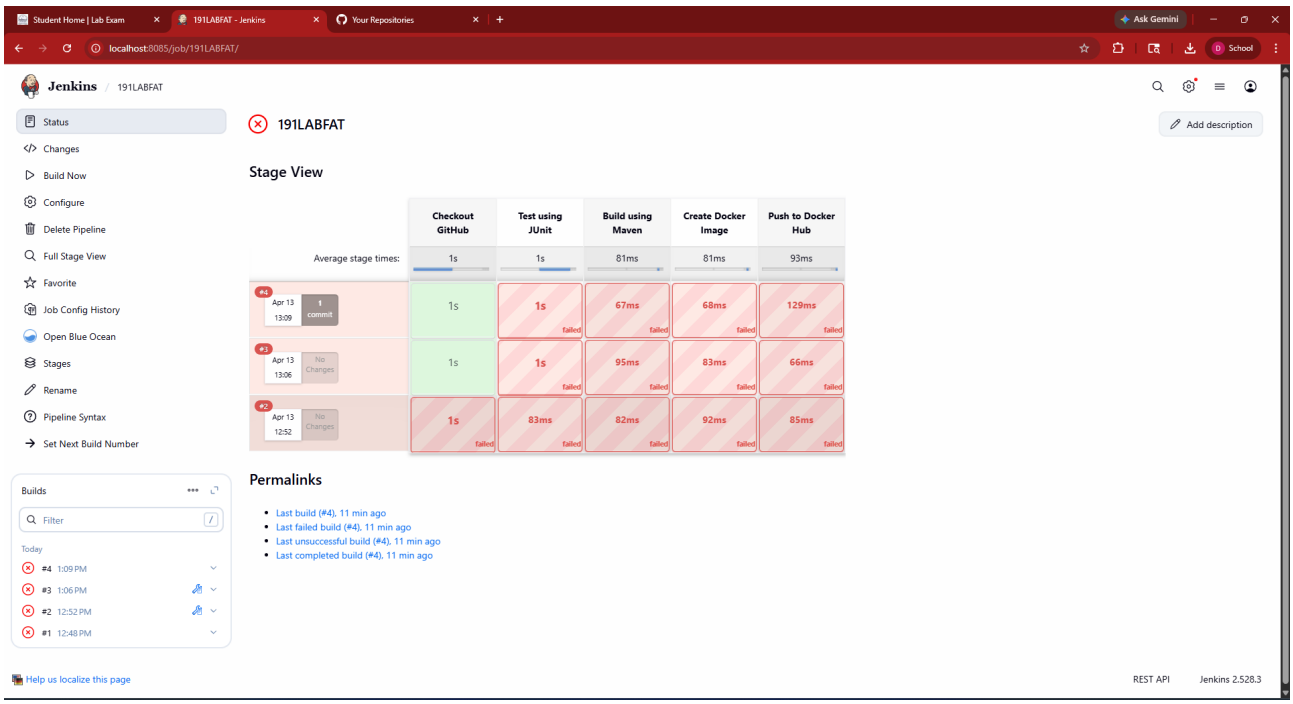

Student Home | Lab Exam
Dashboard - Jenkins
deepakkumars2023/191LABFAT
Ask Gemini
github.com/deepakkumars2023/191LABFAT-AGILE
deepakkumars2023 / 191LABFAT-AGILE
Code Issues Pull requests Actions Projects Wiki Security and quality Insights Settings
On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. Review this update and manage your preferences in your GitHub account settings.
191LABFAT-AGILE Public
Pin Watch 0 Fork 0 Star 0
main 1 Branch 0 Tags Go to file Add file Code
Batch1 and Batch1 first commit cb06492 · 3 minutes ago 1 Commit
src first commit 3 minutes ago
README.md first commit 3 minutes ago
README
191LABFAT-AGILE
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors
No contributors
Languages
Java 100.0%


Student Home | Lab Exam
Dashboard - Jenkins
191LABFAT-AGILE/src/main/jav
Ask Gemini
github.com/deepakkumars2023/191LABFAT-AGILE/blob/main/src/main/java/com/example/Animal.java
School
deepakkumars2023 / 191LABFAT-AGILE
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. Review this update and manage your preferences in your GitHub account settings.
Files
main
Go to file
t
src
main/java/com/example
Animal.java
Dog.java
target
test/java/com/example
pom.xml
README.md
191LABFAT-AGILE / src / main / java / com / example / Animal.java
Batch1 and Batch1 first commit
cb06492 · 5 minutes ago
History
Code
Blame
4 lines (3 loc) · 78 Bytes
Raw
1 public class Animal {
2 public String speak() { return "Animal Sound"; }
3 }


File Edit Selection View Go Run Terminal Help
191
EXPLORER
OPEN EDITORS
Animal.java src\main\java\... 1
DogTest.java src\test\jav... 9+
pom.xml src
Dog.java src\main\java\co... 3
191
src
main\java\com\example
Animal.java 1
Dog.java 3
target
test\java\com\example
DogTest.java 9+
pom.xml
README.md
Animal.java 1
DogTest.java 9+
pom.xml
Dog.java 3
src > main > java > com > example > Animal.java > ...
1 public class Animal {
2 public String speak() { return "Animal Sound"; }
3 }
4
5 //changes made
6
OUTPUT
DEBUG CONSOLE
TERMINAL
PROBLEMS 16
powershell
After doing this, you may fix the identity used for this commit with:
git commit --amend --reset-author
1 file changed, 1 insertion(+)
PS C:\Users\batch1.VITUNIVERSITY\Desktop\191> git push
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 20 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (8/8), 654 bytes | 654.00 KiB/s, done.
Total 8 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/deepakkumars2023/191LABFAT-AGILE.git
cb06492..3fd3d76 main -> main
PS C:\Users\batch1.VITUNIVERSITY\Desktop\191>
OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION
main
16 0
Indexing completed.
Java: Ready
Ln 5, Col 15
Spaces: 4
UTF-8
CRLF
Java
Go Live


Student Home | Lab Exam
Dashboard - Jenkins
191LABFAT-AGILE/src/main/jav
github.com/deepakkumars2023/191LABFAT-AGILE/blob/main/src/main/java/com/example/Animal.java
deepakkumars2023 / 191LABFAT-AGILE
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. Review this update and manage your preferences in your GitHub account settings.
Files
main
Go to file
src
main/java/com/example
Animal.java
Dog.java
target
test
pom.xml
README.md
191LABFAT-AGILE / src / main / java / com / example / Animal.java
Batch1 and Batch1 update
3fd3d76 · now
History
Code
Blame
5 lines (4 loc) · 93 Bytes
Raw
public class Animal {
public String speak() { return "Animal Sound"; }
}
//changes made


Student Home | Lab Exam
191LABFAT Config - Jenkins
191LABFAT-AGILE/src/Dockerfi
localhost:8085/job/191LABFAT/configure
Jenkins / 191LABFAT / Configuration
Configure
General
Office 365 Connector / Power Automate workflows
Triggers
Pipeline
Advanced
Trigger builds remotely (e.g., from scripts)
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script
Script
try sample Pipeline...
stage('Docker Build & Push') {
steps {
script {
// Replace 'your-username' with your actual Docker Hub username
def appImage = docker.build("deepakkumar1241/inheritance-app:${env.BUILD_ID}")
docker.withRegistry('https://docker.io', 'dockerhub-creds') {
appImage.push()
appImage.push("latest")
}
}
}
}
}
}
Use Groovy Sandbox
Pipeline Syntax
Advanced
Save
Apply



https://github.com/deepakkumars2023/191LABFAT-AGILE